Separating the variables,

we get $e^y dy = \left(e^x + x^2\right)dx$

Integrating, we get $\int e^y dy = \int \left(e^x + x^2\right)dx$

$$e^y = e^x + \frac{x^3}{3} + c$$

Example 4.8

Solve $3e^x \tan y dx + (1+e^x)\sec^2 y dy = 0$ given $y(0) = \frac{\pi}{4}$

Solution:

Given $3e^x \tan y dx + (1+e^x)\sec^2 y\ dy = 0$

$$3e^x \tan y\ dx = -(1+e^x)\sec^2 y\ dy$$

$$\frac{3e^x}{1+e^x}dx = -\frac{\sec^2 y}{\tan y}dy$$

Integrating, we get $3\int \frac{e^x}{1+e^x}dx = -\int \frac{\sec^2 y}{\tan y}dy + c$

$$3\log(1+e^x) = -\log\tan y + \log c$$

$$\left[\because \int \frac{f'(x)}{f(x)}dx = \log f(x)\right]$$

$$\log(1+e^x)^3 + \log\tan y = \log c$$

$$\log\left[\left(1+e^x\right)^3 \tan y\right] = \log c$$

$$(1+e^x)^3 \tan y = c \qquad (1)$$

Given $y(0) = \frac{\pi}{4}$ $(i.e)$ $y = \frac{\pi}{4}$ at $x = 0$

$$(1) \Rightarrow (1+e^0)^3 \tan\frac{\pi}{4} = c$$

$$2^3(1) = c$$

$$\Rightarrow c = 8$$

Hence the required solution is $(1+e^x)^3 \tan y = 8$.

Example 4.9

Solve $\sec^2 x \tan y dx + \sec^2 y \tan x dy = 0$

Solution:

Separating the variables, we get

$$\frac{\sec^2 x}{\tan x}dx + \frac{\sec^2 y}{\tan y}dy = 0$$

Integrating, we get

$$\int \frac{\sec^2 x}{\tan x}dx + \int \frac{\sec^2 y}{\tan y}dy = c$$

$$\log\tan x + \log\tan y = \log c$$

$$\log(\tan x \tan y) = \log c$$

$$\tan x\ \tan y = c$$

Example 4.10

Solve $ydx - xdy - 3x^2y^2e^{x^3}dx = 0$.

Solution:

Given equation can be written as $\frac{ydx - xdy}{y^2} - 3x^2e^{x^3}dx = 0$

Integrating, $\int \frac{ydx - xdy}{y^2} - \int 3x^2e^{x^3}dx = c$

$$\int d\left(\frac{x}{y}\right) - \int e^t dt = c$$

(where $t = x^3$ and $dt = 3x^2dx$)

$$\frac{x}{y} - e^t = c$$

$$\frac{x}{y} - e^{x^3} = c$$

Example 4.11

Solve : $x - y\frac{dx}{dy} = a\left(x^2 + \frac{dx}{dy}\right)$

Solution:

Given $x - y\frac{dx}{dy} = a\left(x^2 + \frac{dx}{dy}\right)$

$$x - y\frac{dx}{dy} = ax^2 + a\frac{dx}{dy}$$

$$x - ax^2 = a\frac{dx}{dy} + y\frac{dx}{dy}$$

$$x(1-ax) = (a+y)\frac{dx}{dy}$$

By separating the variables

$$\text{we get, } \frac{dx}{x(1-ax)} = \frac{dy}{a+y}$$

$$\left(\frac{a}{1-ax} + \frac{1}{x}\right)dx = \frac{dy}{a+y}$$

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